

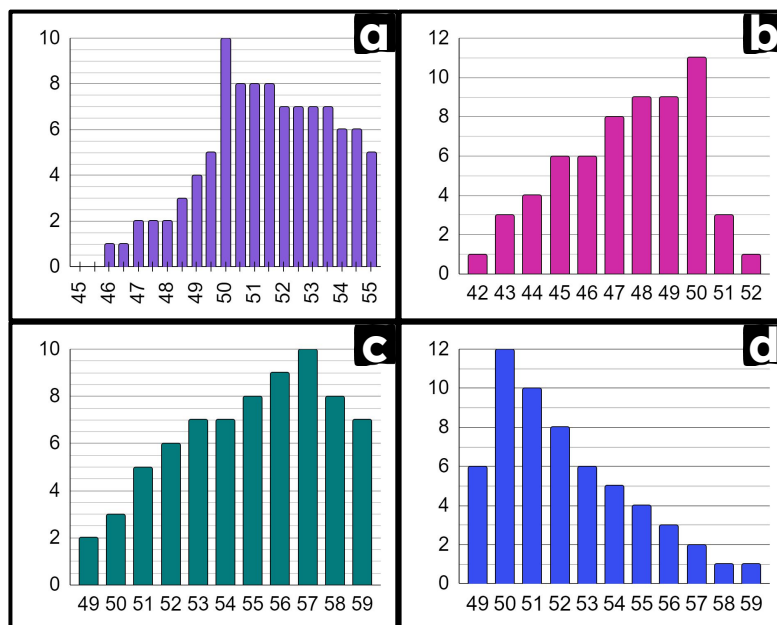


Analysing different statistical representations

Task A: Comparing summary statistics to statistical graphs

1) Which of these summary statistic values is applicable to bar charts a, b, c, d, or any combination of them?

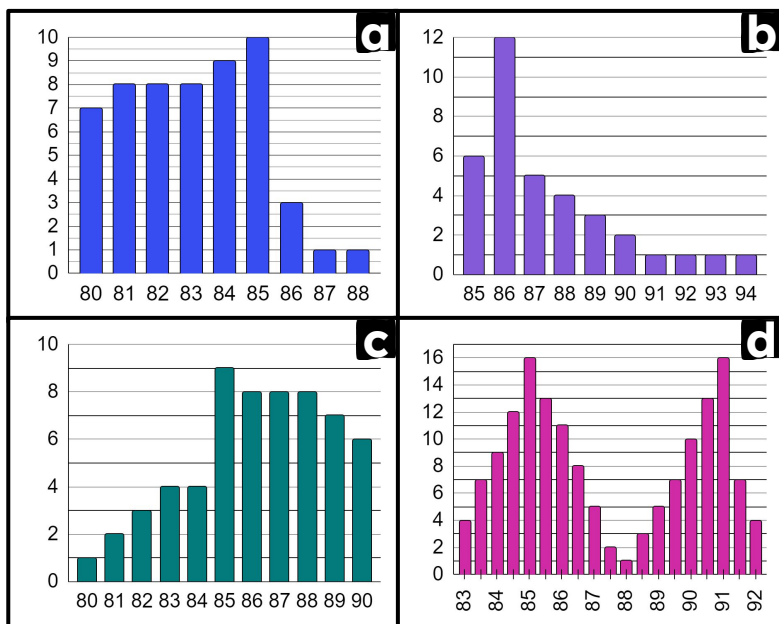
		a	b	c	d
mean	55			✓	
median	48		✓		
mode	50	✓	✓		✓
range	10		✓	✓	✓



2) Each summary statistic below matches with a pair of bar charts. For each summary statistic, write down its value and the names of the two classes it represents.

mean	15	a	c
mean	12.3	b	d
mode	16	a	d
mode	14	c	b
range	10	a	c





3) Place a ✓ by each letter if all summary statistics given accurately represent the corresponding bar chart.

mean		87.4	
a	b	c	d
	✓	✓	✓

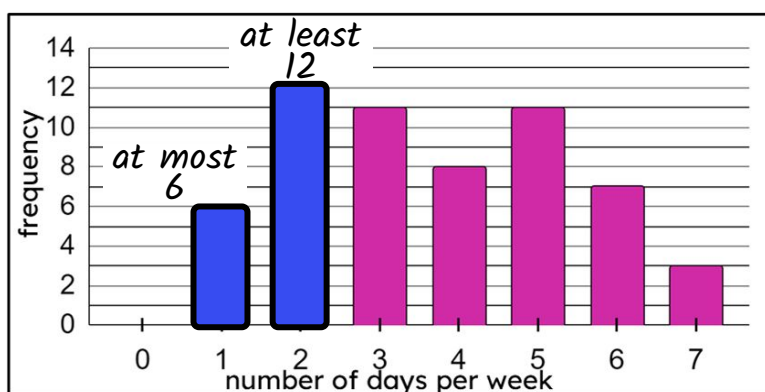
mean		87.4	
range		9	
a	b	c	d
	✓		✓

mean		87.4	
range		9	
mode		85	
a	b	c	d
			✓

Task B: Completing graphs from summary statistics

1) On Monday, Alex goes to his local town centre and collects data on the number of days per week people visit the town centre.

He loses some of the data he collected, but knows that he asked 58 people, the range is 7 days, and the modal number of days is 2.



a) Which bar does Alex know he must leave blank?

0 days

b) How many people are not yet represented on the bar chart?

18 people

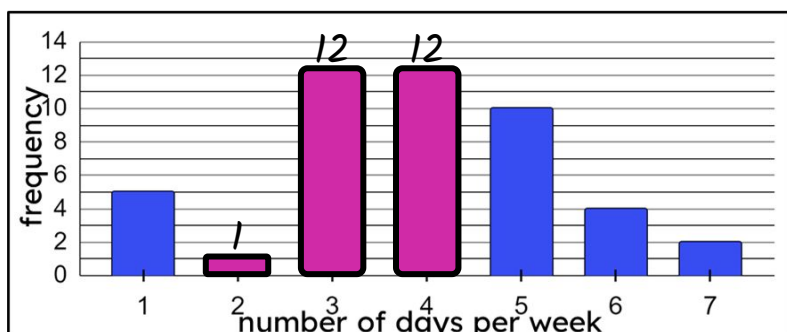
c) Complete a bar chart that shows this.



2) On Saturday, Jun goes to his local town centre and collects data on the number of days per week people visit the town centre.



He loses some of the data he collected, but knows that he asked 46 people, the data are bimodal, and that more people visit 7 times a week than 2 times a week.



a) How many people are not yet represented on the bar chart?

25 people

b) How many people go to town 2 times per week?

1 person

c) Complete the bar chart.

Task C: Sketching graphs from summary statistics

1) Jacob wants to sketch a vertical line chart of these summary statistics.



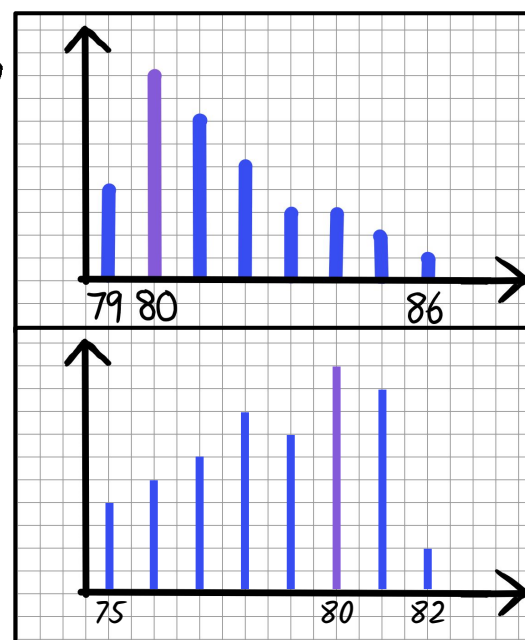
mode	80
range	7

a) Write down a calculation to show why the x-axis must be at least 16 squares across.

$7 \times 2 + 2 = 16$, this is 2 squares per vertical line drawn

b) Complete the vertical line graph that Jacob started.

c) Draw a different vertical line graph that also represents the summary statistics Jacob used.



This is just one example

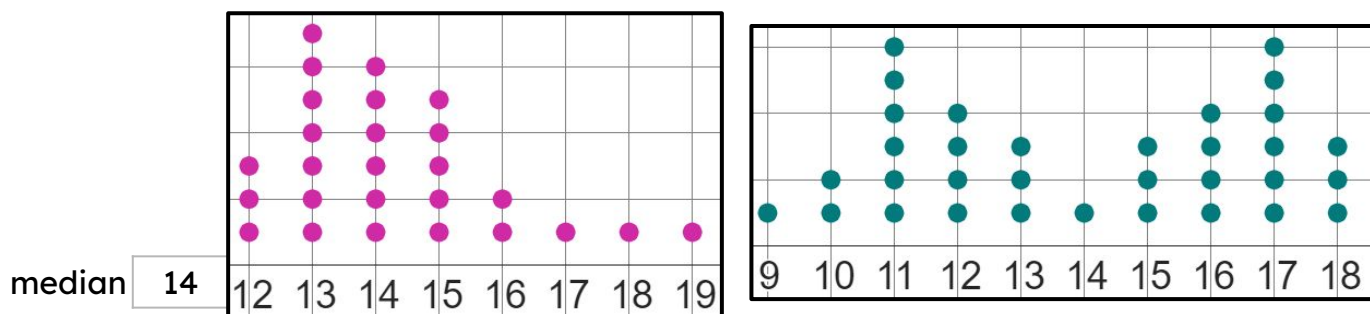


2) Sketch two different dot plots that represents the information below.

One of these dot plots must show a bimodal dataset.

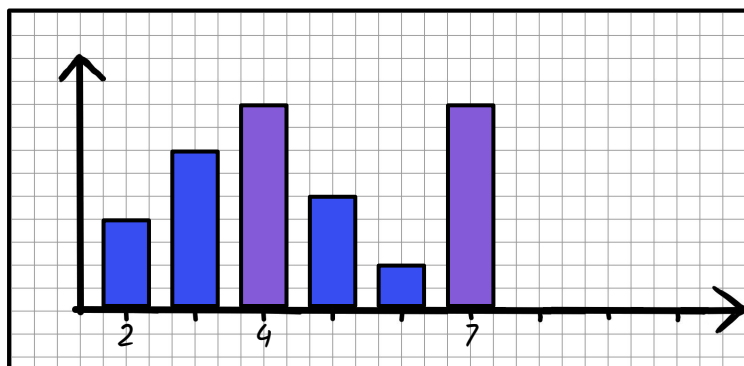
These are examples of the many ways of representing this information on a dot plot.

With the single mode dot plot, 14 should be close to the peak, whilst on the bimodal data, 14 should be in between the two modes.



3) Sketch a bar chart that represents the information below.

mode	bimodal at 4 and 7
range	5
minimum value	2



The bar chart may not go across the whole x-axis.

There are many possible ways of sketching this bar chart.

As long as 2 is the minimum value, 7 is the maximum value, and 4 and 7 are the tallest bars, the rest can be represented differently.